Element K:
The project started with defining the problem that the stakeholders needed to be solved. The issue they faced was the hardships special needs kids had during physical education. Students were unable to participate due to their conditions and the goal of our team was to develop a way that they were able to participate. The team looked at a common game of frisbee and decided on a launcher that requires minimal physical exertion to throw a frisbee.

- This was good in reaching what we thought the stakeholders needed.

The next step was to research prior attempts at frisbee launchers. We looked at 4 different takes on the idea and analyzed how each one worked, along with their pros and cons.

- Each of the attempts were useful in giving what worked well, along with comparing them to our restraints.

After research on prior attempts, design requirements were laid out to what the product should be able to reach. We looked at what the prioritizes of what the stakeholders wanted to create a guideline around them. A list of baseline requirements were created to detail areas of focus.

- A list of requirements were very helpful in deciding what should be prioritized more when generating future ideas, but they were basic at their core and could've been more specific.

Using the prior attempts and the guidelines as inspiration, different launch methods were generated and compared in a design matrix. The highest scoring design then had a layout of materials, prices, and why the part is included for future reference. A 3d model was then created along with a plan of action to create a prototype.

- The early designs served as a barebone design to decide which method was the most efficient by our standards, but the final design was not further changed from the earlier designs. This may have made the final product unclean as it was not tuned finely.

At this point, the project was handed off to an expert to review STEM application and provide feedback

- There was, however, only one expert that reviewed this project. This likely caused a lack of multiple opinions to draw from and solely relied on that one expert.

The final design was further judged on how well they met the stakeholder's needs. Justifications on the viability and importance of matching the initial problem statement then needed to be written up with reason.

- The justifications matched what the stakeholders needed, but only that. It didn't plan to extend further than the minimum of reaching their goals.

The final product was a curved launcher using a flywheel to generate power and friction to force a spin out the frisbee. The wall was adjustable to move inwards to allow adjustability of a smaller frisbee. The construction process started with gathering materials. The general shape sketched on the base piece and alignment of the motor and walls were sketched on from the 3d model. Attaching the wall to allow for adjustability meant the walls had to be removable, but the adjustability was later scrapped due to time. Motor testing with the pressure on the frisbee was then later conducted after the base piece was cut out and walls installed.

- Time constraints left the product messy and the adjustable walls ideas were scrapped. The overall appearance was messy due to it being unable to be touched up in the time span, along with edges being sharp and other areas being rough (very unsuitable for younger students).

Testing plans were constructed around high priorities of the stakeholders. This left distance, accuracy, and size to be key areas of focus for testing. Goal lines were set for each of the requirements in pass and fail formatting.

- Areas outside of the ones listed were not looked at all (safety, battery power, etc). This was due to time constraints. This led to an unpolished product and couldn't be easily iterated upon due to a lack of information of what to improve.